Amin Weinman

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Education

New Mexico Institute of Mining and Technology Expected: May 2027

- BS in Computer Science, Minor in Mathematics, GPA: 3.5/4.0
- **Coursework:** Programming Languages, Data Structures, Computer Architecture, Machine Learning, Statistics

Experience

Researcher, DREU in Agentic AI – Daytona Beach, FL May 2026 – August 2026

- Designed and evaluated a guarded, tool-using LLM agent (5 tools, 5-guardrail stack) for anomaly triage; evaluated 8 agent configurations across 480 adversarial test cases.
- Designed a guardrail ablation study cutting prompt-injection success from 100% to 0% and raising safe-behavior compliance from 22% to 100%.
- Built an anomaly-detection pipeline (Isolation Forest, z-score baselines) over 142K+ satellite telemetry rows; found and fixed an AUC evaluation bug inflating results to 0.98, correcting to 0.73.

Software Engineer Intern, Metrical.al – San Francisco, CA June 2025 – May 2026

- Architected high-performance distributed systems with modular architecture, improving system reliability by 40% for enterprise-scale applications
- Developed comprehensive full-stack solution supporting CLI, REST API, and web interfaces, optimizing for sub-50ms latency performance
- Built and evaluated a promo-response modeling workflow that improved checkout-targeting performance by 0.135 AUC through feature engineering and model comparison.

Software Engineer Intern, Sandia National Labs – Albuquerque, NM May 2024 – May 2025

- Implemented high-performance messaging systems within distributed cloud infrastructure, reducing data processing latency by 35% for cybersecurity applications
- Developed Python algorithms leveraging advanced data structures and optimized computer architecture, achieving 10x throughput improvement for high-volume data processing
- Engineered API integration systems connecting Large Language Models, enabling seamless cross-platform communication for 5+ development teams

Projects

Client Library Harness July 2026

- Built and maintain a client-side demo/test harness, used internally to validate and showcase tracking, personalization, and offer-flow behavior across real e-commerce interaction patterns.
- Led a migration from legacy jQuery integration pages to a React/Vite SPA that simulates a full purchase funnel (browse, cart, checkout, confirmation) with shared cart state, inline validation, and environment-driven config, deployed independently to dev, stage, and production.

xPTS Shot Quality Model GitHub | May 2026

- Developed an NBA shot quality model that estimates expected points for each shot attempt using contextual features such as shot distance, angle, game clock, and pressure.
- Combined together the full analytics workflow in one reproducible project, including shot data collection, feature engineering, baseline modeling, evaluation metrics, and visual outputs such as shot charts and calibration views.

Premier League Predictor Demo | August 2026

- Built a Premier League predictor combining a Dixon-Coles time-decayed Poisson model with Monte Carlo simulation, validated via 3-season backtesting against closing betting-market odds.
- Implemented Quantum Amplitude Estimation (Qiskit) on a real model probability, demonstrating QAE's theoretical quadratic query-complexity speedup over classical Monte Carlo sampling.
- Evaluated 9 candidate model improvements via rigorous backtest-driven A/B testing, keeping only statistically validated gains and documenting rejected hypotheses.

Technical Skills

Languages: Python, C, JavaScript, SQL

Frameworks & Tools: React, Flask, PostgreSQL, Git, Docker, REST APIs, TensorFlow, Qiskit

Technical Areas: Distributed Systems, Machine Learning, Algorithms, Data Structures